

KIWI Primer 20K FPGA BOARD

Gowin FPGA 20K LUTs

KIWI Primer 20K FPGA USER GUIDE

15/11/2025

KIWI Primer 20K FPGA BOARD

Gowin FPGA 20K LUTs

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Revision History

Date	Version	Description
01/05/2025	1.0	Research and test
15/11/2025	1.1	Initial version published

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1 About This Guide

1.1 Purpose

The Kiwi Primer 20k V1.1 development board user manual consists of the following four parts:

- A brief introduction to the features and hardware resources of the development board
- An introduction to the hardware circuits functions, circuit, and pinout
- Considerations for the use of the development board
- Introduction to the use of the FPGA development software.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI Website. You can find the related documents at www.gowinsemi.com

- [GW2A series of FPGA Products](#) ,datasheet
- <https://cdn.gowinsemi.com.cn/UG110E.pdf> ,Pinout
- <https://cdn.gowinsemi.com.cn/UG111E.pdf> , GW2A series of FPGA Products Package and Pinout
- <https://cdn.gowinsemi.com.cn/UG290E.pdf> ,Gowin Programming and Configuration Guide
- <https://cdn.gowinsemi.com.cn/SUG100E.pdf> , Gowin software user guide

1.3 Abbreviations and Terminology

Abbreviations and Terminology	Full Name
BSRAM	Block SRAM
DLL	Delay-locked Loop
DSP	Digital Signal Processing
FPGA	Field Programmable Gate Array
GPIO	General Purpose Input Output
LDO	Low Dropout Regulator
LED	Light Emitting Diode
LQ100	LQFP100
LUT4	Four-input Look-up Table
PLL	Phase-locked Loop
SSRAM	Shadow SRAM

Table 1 Table Of abbreviations and terminology

1.4 Support and Feedback

Onekiwi Technology provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

- Website: <https://onekiwi.com.vn/>
- Email: support@ontech.com.vn

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2 Development Board Description

2.1 Overview



Figure 2-1 Front view of Development board



Figure 2-2 Back view of Development board

Onekiwi Primer 20K development board applies to high speed data storage, high-speed communication test, FPGA functions evaluation, the verification of hardware reliability, software learning and debugging.

The development board uses the GW2A-LV18PG256 FPGA device, which is the first generation product of Gowin Arora family. The GW2A series of FPGA products offer a range of comprehensive features and rich internal resources like high-performance DSP resources, a high-speed LVDS interface, and abundant BSRAM memory resources. These embedded resources combine a streamlined FPGA architecture with a 55nm process to make the GW2A series of FPGA products ideal for high-speed and low-cost applications.

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2.2 A Development Board Suite

A development board suite includes the following items:

- Onekiwi GW2A-LV18PG256-V1.1 development board
- USB type C

1



1. Development board
2. USB cable type C

2



Figure 2-3 Development board suite

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2.3 PCB components

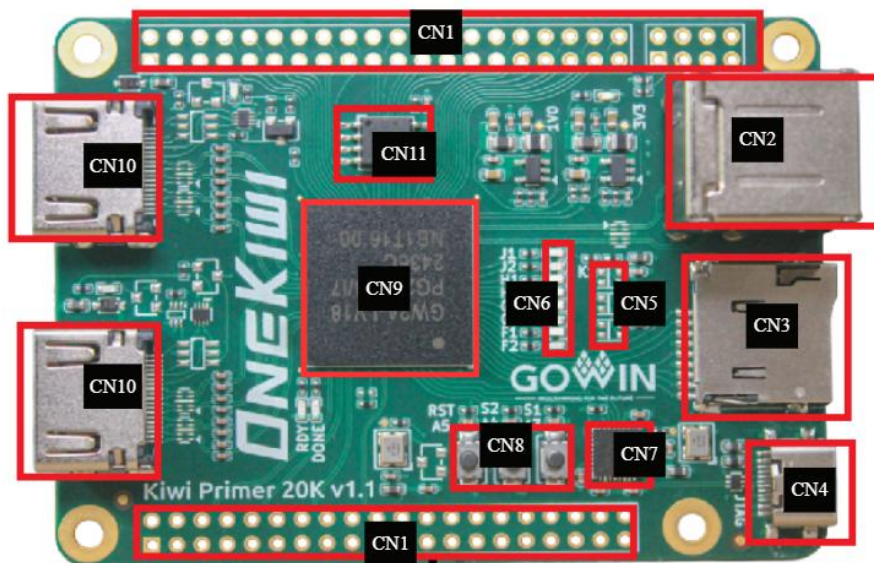


Figure 2-4 Onekiwi nano 4k Components

CN	Name
CN1	GPIOs Pin header
CN2	USB 2.0
CN3	Micro SD card
CN4	JTAG USB debug
CN5	LED RGB
CN6	LED
CN7	Gwu2x IC
CN8	Button
CN9	FPGA Chip
CN10	HDMI
CN11	Configuration Flash

Table 2 Onekiwi 20k components

2.4 System Architecture

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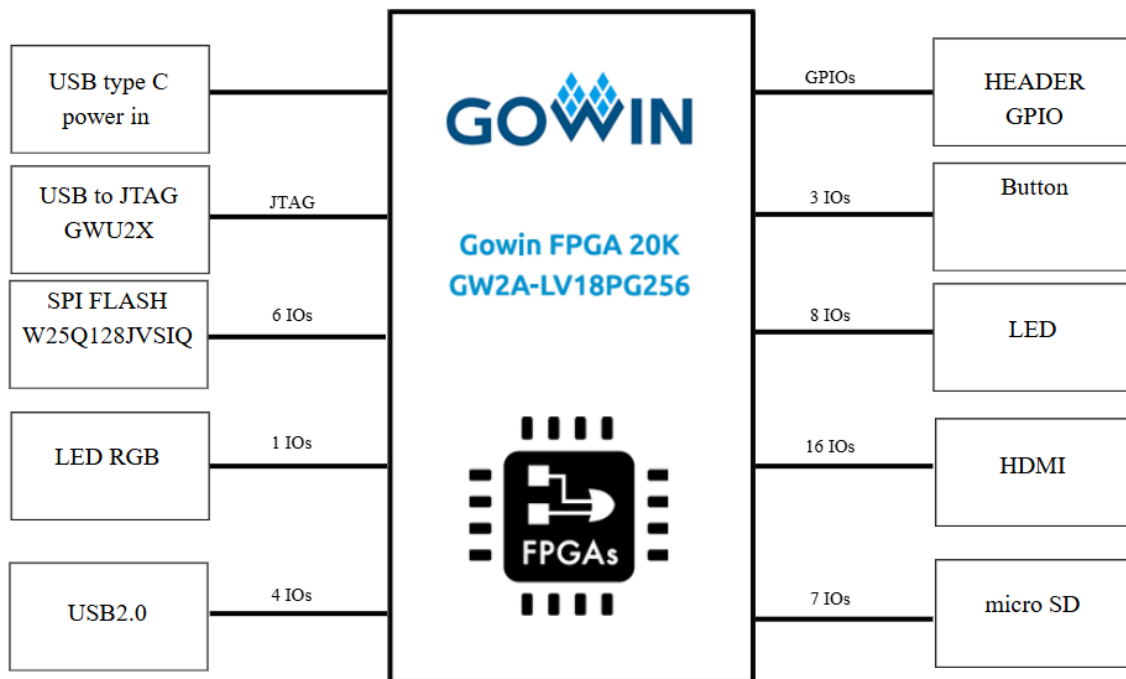


Figure 2-5 System architecture

2.5 Features

- **FPGA CORE**
 - GW2A-LVPBG256C8/I7 FPGA
 - Built-in Flash,
 - LUT4 + BSRAM
 - GPIOs, LVDSs
- **POWER SYSTEM**
 - 5V Input
 - 3.3V (I/O, interface)
- **CLOCK**
 - 27 MHz Oscillator → FPGA
 - 12 MHz Oscillator → USB GWU2X Interface
- **USB INTERFACE**
 - GWU2X chip
 - USB-C Connector
- **GPIO HEADERS**
 - 40x2 and 8 pin headers
 - 70+ GPIOs
- **USER INPUT + OUTPUT**
 - SD Card
 - USB 2.0

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- HDMI
- 3× Keys
 - Reset
 - 2x User
- 3x Led RGB
- 9× LEDs
 - Power
 - 8× User

- **SPI FLASH**
- 128Mbit MSPI Flash for config

- **JTAG**
- TCK, TMS, TDI, TDO via header

2.6 Development Board Specification

Item	Specification
FPGA Chip	GW2A-LV18PG256 (Gowin Arora series)
Logic Units (LUT4)	20,736
Registers (Flip-flops)	15,552
Block SRAM (total)	828 Kbits
Shadow SRAM	41,472 bits
18×18 Multipliers	48
Number of PLLs	4
Number of I/O Banks	8
On-board Memory / Flash	Flash 64 Mbit
Available User I/O (approx.)	~70+ user-accessible I/O
Main Voltage / Power Supply	5 V (via USB or main power input)
Operating Ambient Temperature	−10 °C to +65 °C
Typical Use	Development board for learning FPGA, IP verification, and embedded control applications

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3 FPGA Circuits

3.1 FPGA Module

3.2 Download Interface

3.2.1 Overview

The development board provides an USB download interface. The data stream file can be downloaded to the internal SRAM, internal flash, or external flash as needed.

Note!

- When downloaded to SRAM, the data stream file will be lost if the device is power down, and it will need to be downloaded again after power-on.
- If downloaded to flash, the data stream file will not be lost if the device is powered down.

3.2.2 USB Download Circuit

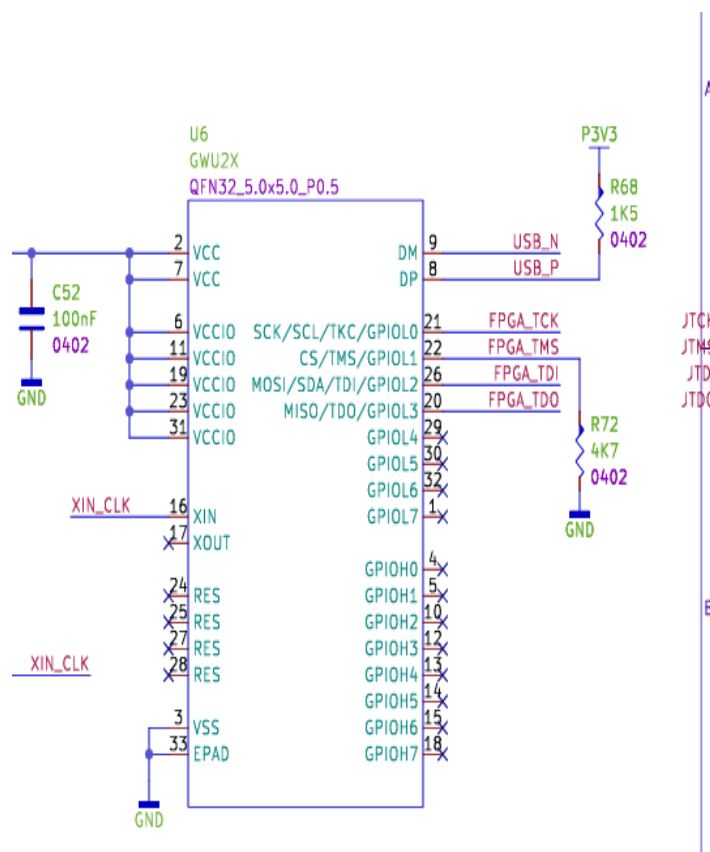


Figure 3-1 GWU2X USB TO JTAG Chip

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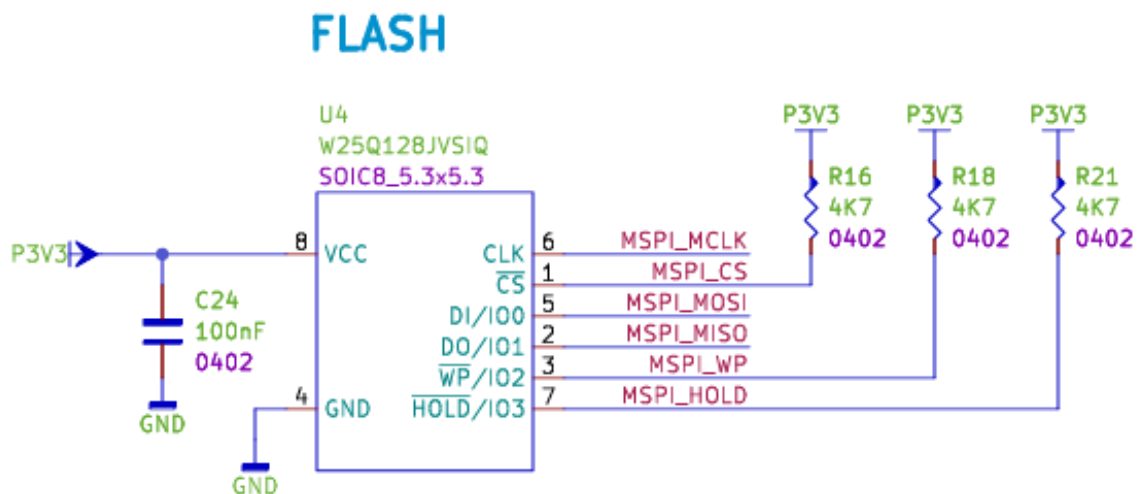


Figure 3-2 SPI flash downloading

3.2.3 Downloading the Data Stream

The data stream file can be downloaded in the following ways.

1. SRAM

Scan the device and download the bit file after powering the device on. The Done indicator lights up to denote that the download has been successful.

2. Internal Flash

Power on and download. After downloading the data stream file successfully, power down to reset and load the bit file from the internal Flash, and when the Done indicator lights up to denote that the download has been successful.

3. External Flash Configuration

Power on and download. Power down to reset and load the bit file from the external flash. The Done indicator lights up to denote that the download has been successful.

3.2.4 Pinout

Signal Name	Pin No.	BANK	Description	I/O
FPGA_TMS	B8	1	JTAG Signal	VCCIO1
FPGA_TCK	A7	1	JTAG Signal	VCCIO1
FPGA_TDI	A6	1	JTAG Signal	VCCIO1
FPGA_TDO	C6	1	JTAG Signal	VCCIO1
MSPI_MCLK	L10	3	FLASH signals configuration	VCCIO3
MSPI_CS	M9	3	FLASH signals configuration	VCCIO3
MSPI_WP			FLASH signals configuration	
MSPI_HOLD			FLASH signals configuration	
MSPI_MISO	P10	3	FLASH signals configuration	VCCIO3
MSPI_MOSI	R10	3	FLASH signals configuration	VCCIO3

Table 3 Pinout Downloading

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3.3 Power Supply

3.3.1 Overview

The 5V input power interface has overcurrent and inverse current protection. The overcurrent limit is 2A.

The TLV6 power supply chip is used to step down voltage from 5V to 3.3V, The power supply can support up to 2A, which can meet the power demand of the development board.

3.3.2 Power System Distribution

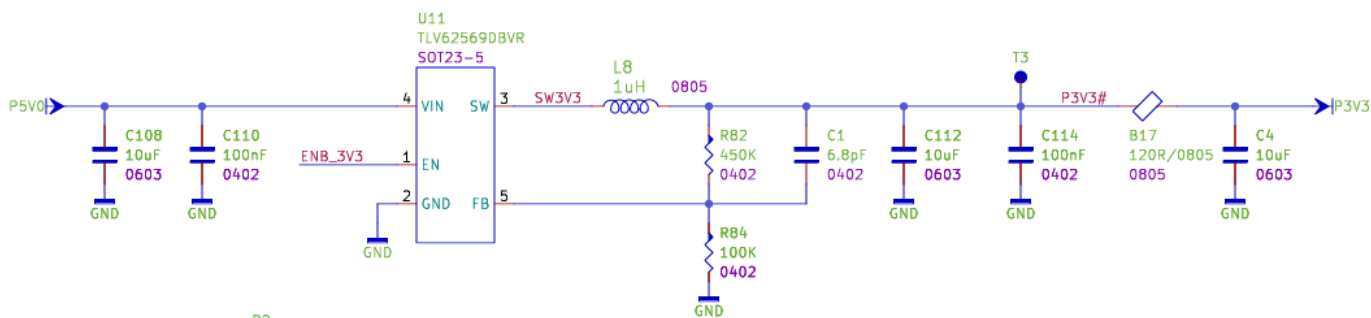


Figure 3-3 Power distribution

3.4 Clock

3.4.1 Overview

A 27MHz crystal oscillator is provided in the development board that connects to the PLL input pin. This can be employed as the input clock for the PLL in FPGA, and the output clock as needed via multiplication and division of the PLL frequency.

To facilitate testing, a SMA socket is reserved on the development board as the clock input interface. The clock signal is connected to the FPGA global clock pin.

3.4.2 Clock

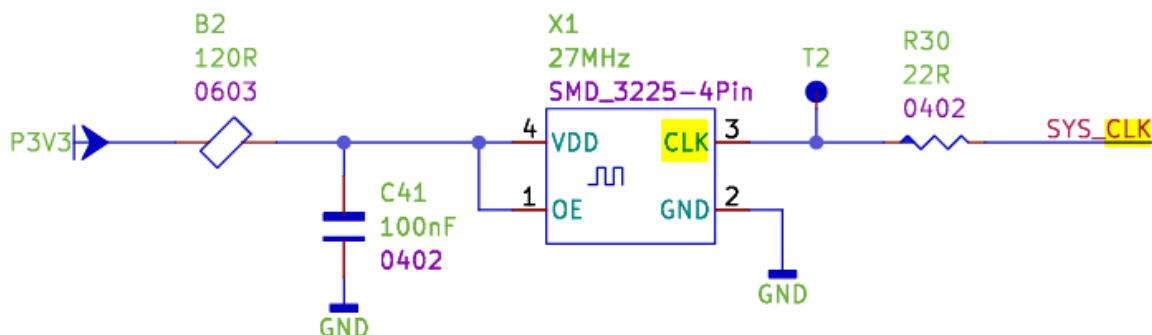


Figure 3-4 Clock circuit

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3.4.3 Pinout

Signal Name	Pin No.	BANK	Description	I/O
SYS_CLK	H11	0	27 MHz crystal oscillator input	3.3V

Table 4 Pinout Clock

3.5 LED

3.5.1 Overview

Seven green LEDs are incorporated into the development board and are used to display the required status. In addition, Seven LEDs are reserved to signify power supply and FPGA loading status.

Users can test the LEDs in the following ways:

- If the output signal of related pins is logic low, LED is on.
- If logic is high, LED is off.

3.5.2 LED Circuit

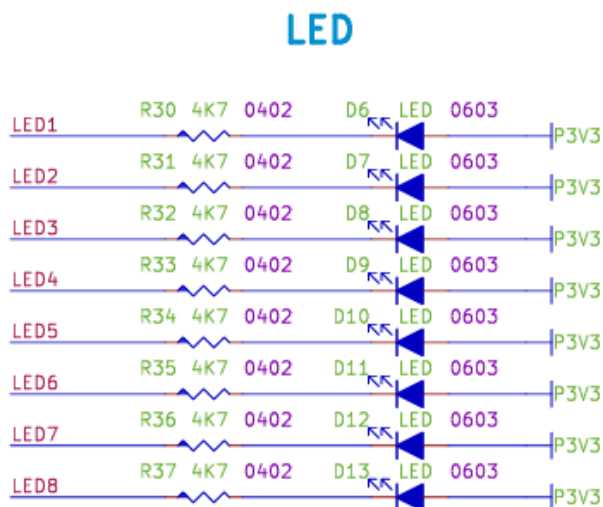


Figure 3-5 Led circuit

3.5.3 Pinout

Signal Name	Pin No.	BANK	Description	I/O
LED1	J1	5	LED1	3.3V
LED2	J2	4	LED2	3.3V
LED3	H1	4	LED3	3.3V
LED4	H2	5	LED4	3.3V
LED5	G1	5	LED5	3.3V
LED6	G2	5	LED6	3.3V
LED7	F1	5	LED7	3.3V

Table 5 Led pinout

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3.6 Key

3.6.1 Overview

Two key switches are embedded in the development board. Users can manually input a low level to the corresponding FPGA pins for testing purposes.

3.6.2 Key Circuit

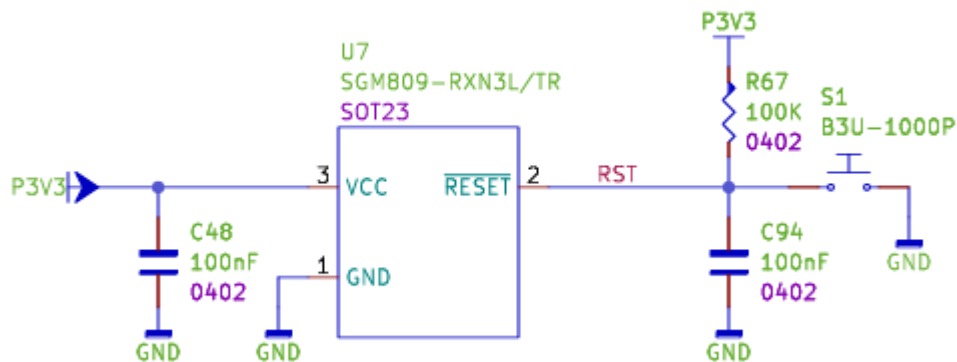


Figure 3-6 Key reset circuit

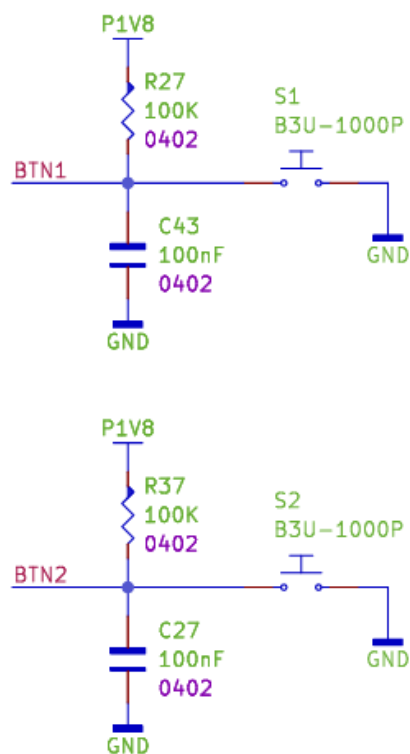


Figure 3-7 Key circuit

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3.6.3 Pinout

Signal Name	Pin No.	BANK	Description	I/O
BTN1	A3	5	KEY1	3.3V
BTN2	A4	5	KEY2	3.3V
RST	A5	5	RESET	3.3V

Table 6 Key pinout

3.7 HDMI

3.7.1 Overview

The Kiwi primer 20K board integrates a high-performance HDMI interfaces designed to support high-resolution video output for display, multimedia, and embedded imaging applications. The board has two HDMI displays.

3.7.2 HDMI circuit

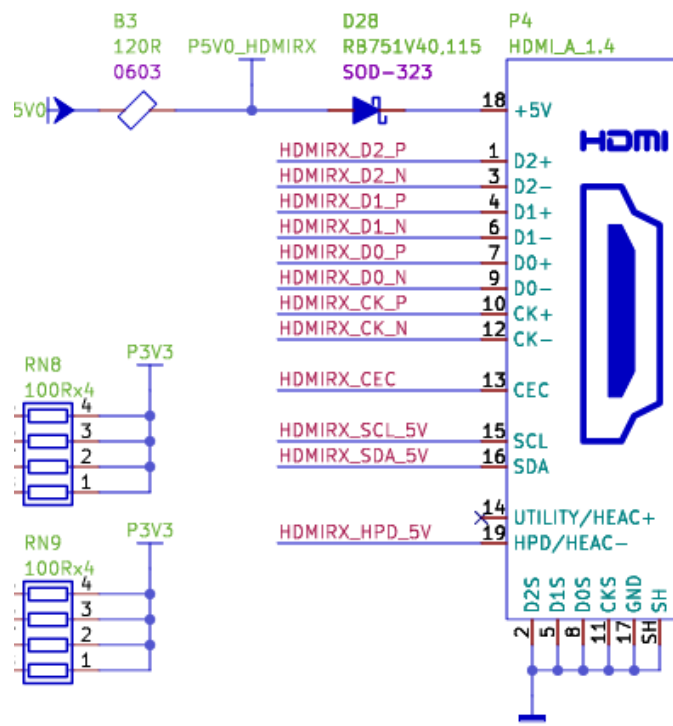


Figure 3-8 HDMI RX circuit

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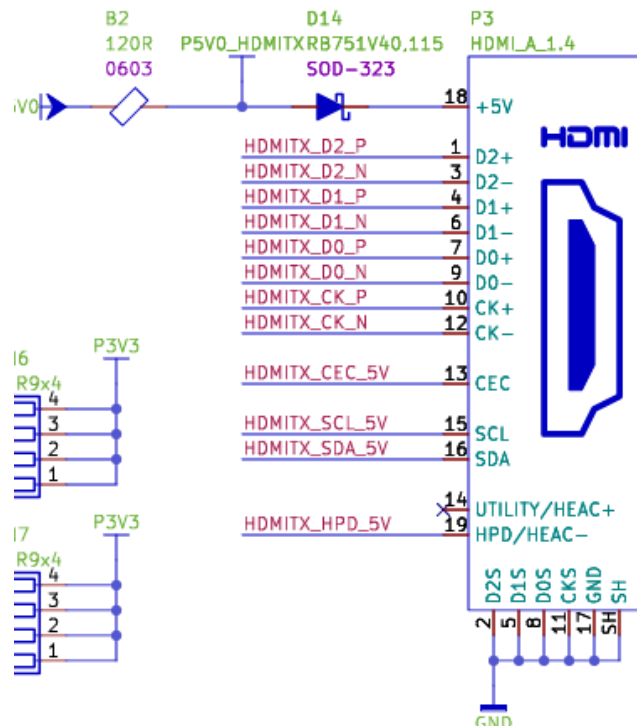


Figure 3-9 HDMI TX circuit

3.7.3 Pinout

Signal Name	Pin No.	BANK	I/O
HDMITX_D2_P	H13	0	VCCIO0
HDMITX_D2_N	J12	0	VCCIO0
HDMITX_D1_P	H14	0	VCCIO0
HDMITX_D1_N	H16	0	VCCIO0
HDMITX_D0_P	J16	0	VCCIO0
HDMITX_D0_N	J14	0	VCCIO0
HDMITX_CK_P	J15	0	VCCIO0
HDMITX_CK_N	K16	0	VCCIO0
HDMIRX_D2_P	D16	0	VCCIO0
HDMIRX_D2_N	E14	0	VCCIO0
HDMIRX_D1_P	E16	0	VCCIO0
HDMIRX_D1_N	F15	0	VCCIO0
HDMIRX_D0_P	F13	0	VCCIO0
HDMIRX_D0_N	G12	0	VCCIO0
HDMIRX_CK_P	G16	0	VCCIO0
HDMIRX_CK_N	H15	0	VCCIO0

Table 7 HDMI Pinout

3.8 SD card

3.8.1 Overview

The Kiwi Primer 20K integrates a microSD card interface designed to provide external non-volatile storage for FPGA

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applications. The SD card module allows the system to store and access configuration data, multimedia files, firmware images, logging information, or user-defined data. It communicates with the FPGA through a standard SPI-based SD interface, ensuring simple implementation and compatibility with common IP cores.

3.8.2 Camera Circuit

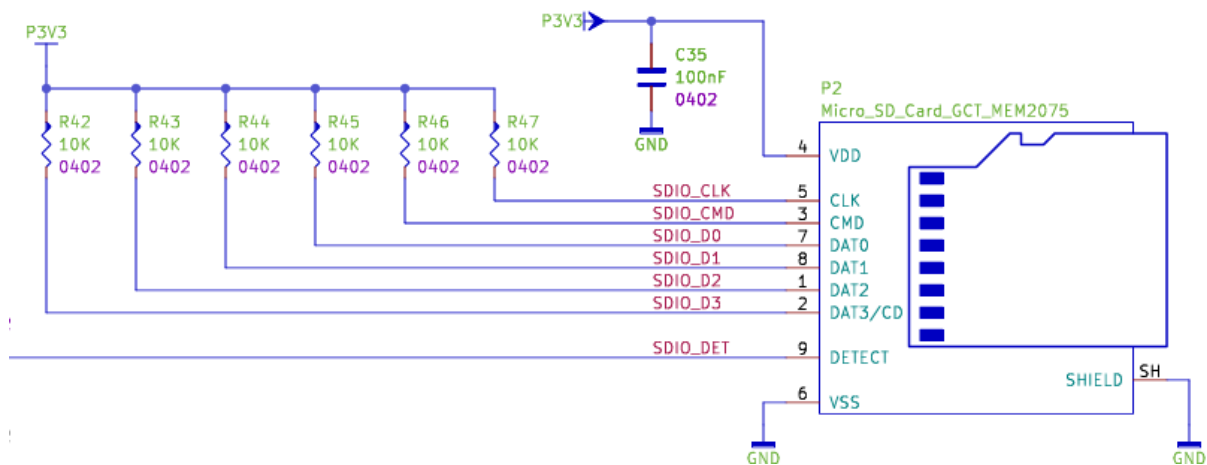


Figure 3-10 Camera circuit

3.8.3 Pinout

Signal Name	Pin No.	BANK	I/O
SDIO_CLK	D2	5	VCCIO5
SDIO_CMD	D1	5	VCCIO5
SDIO_D0	C1	5	VCCIO5
SDIO_D1	C2	5	VCCIO5
SDIO_D2	E1	5	VCCIO5
SDIO_D3	E2	5	VCCIO5
SDIO_DET	B1	5	VCCIO5

Table 8 SD card Pinout

3.9 USB 2.0

3.9.1 Overview

The OneKiwi Primer 20K features a USB 2.0 interface that enables direct connection to common external peripherals such as keyboards, mice, and game controllers. This interface allows the FPGA to communicate with USB-HID devices, making it suitable for human-machine interaction, user input processing, and embedded control applications.

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3.9.2 USB circuit

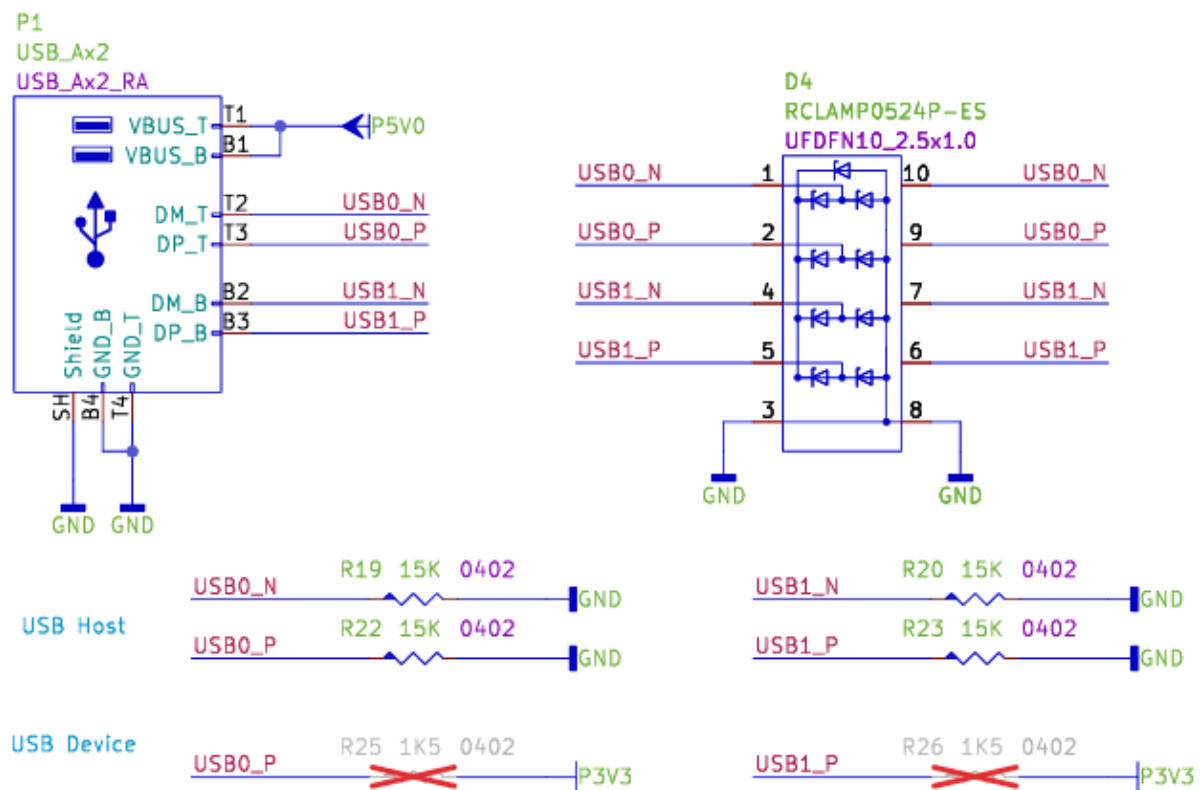


Figure 3-11 USB 2.0 circuit

3.9.3 Pinout

Signal Name	Pin No.	BANK	I/O
USB0_P	N2	4	VCCIO4
USB0_N	P1	4	VCCIO4
USB1_P	M3	4	VCCIO4
USB1_N	N1	4	VCCIO4

Table 9 USB Pinout

3.10 RGB Led

3.10.1 Overview

The OneKiwi Primer 20K integrates an onboard RGB LED that provides a simple and effective visual indicator for status monitoring, debugging, and user feedback. The RGB LED consists of three independent color channels—Red, Green, and Blue—each driven directly by FPGA I/O pins. By adjusting the output logic or PWM signals from the FPGA, the LED can produce a wide range of colors and brightness levels.

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3.10.2 RGB Circuit

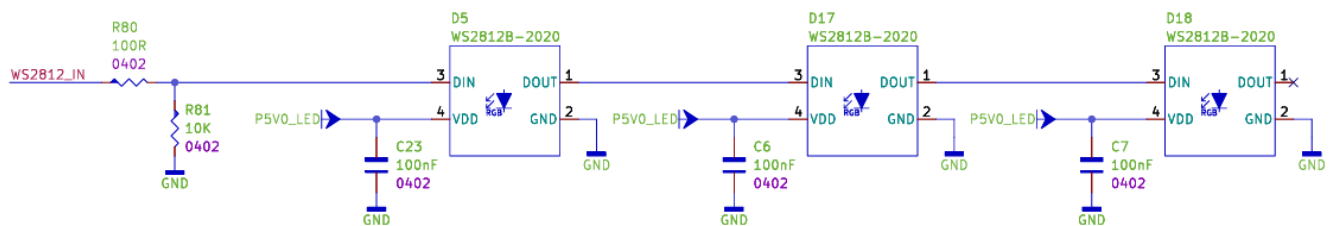


Figure 3-12 RGB Circuit

3.10.3 Pinout

Signal Name	Pin No.	BANK	Description	I/O
WS2812_IN	K1	4	3 LED RGB	3.3V

Table 10 RGB Pinout

3.11 GPIO

3.11.1 Overview

Two 2.54mm 40P and 8P sockets are reserved on the development board for user function extension and testing purposes.

3.11.2 GPIO Circuit

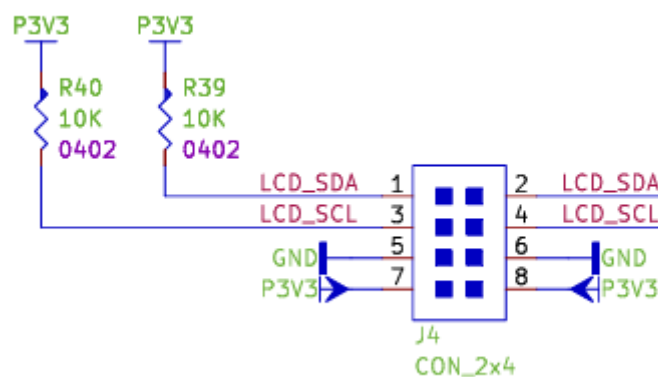


Figure 3-13 8P sockets Circuit

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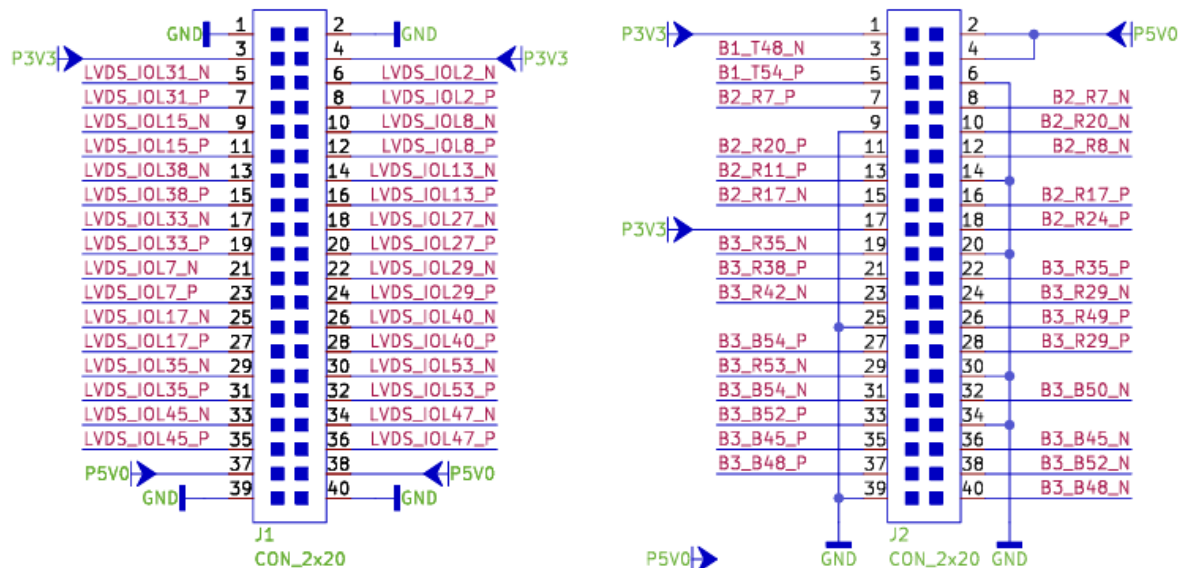


Figure 3-14 40P sockers circuit

3.11.3 Pinout

Signal Name	Pin No.	Socket Pin No.	BANK	Description	I/O
LCD_SDA	P2	1	4	General I/O	
LCD_SDA	P2	2	4	General I/O	
LCD_SCL	R1	3	4	General I/O	
LCD_SCL	R1	4	4	General I/O	
GND		5		Ground	
GND		6		Ground	
3.3V		7		Power	3.3V
3.3V		8		Power	3.3V

Table 11 8P socker Pinout

Signal Name	Pin No.	Socket Pin No.	BANK	Description	I/O
GND	—	1	—	Ground	—
GND	—	2	—	Ground	—
P3V3	—	3	—	Power	
P3V3	—	4	—	Power	
LVDS_IOL31_N	F9	5	6	LVDS	—
LVDS_IOL2_N	A15	6	7	LVDS	—
LVDS_IOL31_P	F9	7	6	LVDS	—
LVDS_IOL2_P	B14	8	7	LVDS	—
LVDS_IOL15_N	C11	9	7	LVDS	—
LVDS_IOL8_N	A14	10	7	LVDS	—
LVDS_IOL15_P	A11	11	7	LVDS	—
LVDS_IOL8_P	B13	12	7	LVDS	—
LVDS_IOL38_N	E9	13	6	LVDS	—

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LVDS_IOL13_N	A12	14	7	LVDS	—
LVDS_IOL38_P	D8	15	6	LVDS	—
LVDS_IOL13_P	B11	16	7	LVDS	—
LVDS_IOL33_N	A10	17	6	LVDS	—
LVDS_IOL27_N	C9	18	7	LVDS	—
LVDS_IOL33_P	B9	19	6	LVDS	—
LVDS_IOL27_P	A9	20	7	LVDS	—
LVDS_IOL7_N	B12	21	7	LVDS	—
LVDS_IOL29_N	A8	22	6	LVDS	—
LVDS_IOL7_P	C12	23	7	LVDS	—
LVDS_IOL29_P	C8	24	6	LVDS	—
LVDS_IOL17_N	E10	25	7	LVDS	—
LVDS_IOL40_N	C7	26	6	LVDS	—
LVDS_IOL17_P	D10	27	7	LVDS	—
LVDS_IOL40_P	B7	28	6	LVDS	—
LVDS_IOL35_N	D9	29	6	LVDS	—
LVDS_IOL53_N	D7	30	6	LVDS	—
LVDS_IOL35_P	F8	31	6	LVDS	—
LVDS_IOL53_P	E6	32	6	LVDS	—
LVDS_IOL45_N	E8	33	6	LVDS	—
LVDS_IOL47_N	B5	34	6	LVDS	—
LVDS_IOL45_P	F7	35	6	LVDS	—
LVDS_IOL47_P	C4	36	6	LVDS	—
P5V0	—	37	—	Power	5V
P5V0	—	38	—	Power	5V
GND	—	39	—	Ground	—
GND	—	40	—	Ground	—

Table 12 40P J1 Pinout

Signal Name	Pin No.	Socket Pin No.	BANK	Description	I/O
P3V3	—	1	—	Power	3.3V
P5V0	—	2	—	Power	5V
B1_T48_N	P16	3	1	General I/O	—
P5V0	—	4	—	Power	5V
B1_T54_P	P15	5	1	General I/O	—
GND	—	6	—	Ground	—
B2_R7_P	T15	7	2	General I/O	—
B2_R7_N	R14	8	2	General I/O	—
GND	—	9	—	Ground	—
B2_R20_N	T14	10	2	General I/O	—
B2_R20_P	R13	11		General I/O	—
B2_R8_N	T13	12	2	General I/O	—
B2_R11_p	R12	13	2	General I/O	—
GND	—	14	—	Ground	—
B2_R17_N	T12	15	2	General I/O	—
B2_R17_P	R11	16	2	General I/O	—
P3V3	—	17	—	Power	3.3V
B3_R24_P	T11	18	2	General I/O	—
B3_R35_N	T10	19	3	General I/O	—
GND	—	20	—	Ground	—
B3_R38_P	T9	21	3	General I/O	—
B3_R35_P	R10	22	3	General I/O	—
B3_R42_N	T8	23	3	General I/O	—
B3_R29_N	R8	24	3	General I/O	—

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GND	—	25	—	Ground	—
B3_R49_P	R7	26	3	General I/O	—
B3_B54_P	R6	27	4	General I/O	—
B3_R29_P	T7	28	3	General I/O	—
B3_R53_N	T6	29	3	General I/O	—
GND	—	30	—	Ground	—
B3_B54_N	T5	31	4	General I/O	—
B3_B50_N	R5	32	4	General I/O	—
B3_B52_P	R4	33	4	General I/O	—
GND	—	34	—	Ground	—
B3_B45_P	P4	35	4	General I/O	—
B3_B45_N	T4	36	4	General I/O	—
B3_B48_P	R3	37	4	General I/O	—
B3_B52_N	T3	38	—	General I/O	—
GND	—	39	—	Ground	—
B3_B48_N	T2	40	4	Ground	—

Table 13 40P J2 Pinout

4 Consideration

Considerations for the use of the development board:

1. Handle with care and pay attention to electrostatic protection.
2. When downloading bitstream files to internal flash or external flash, set the MODE pin state to the correct configuration value, please refer to: <https://cdn.gowinsemi.com.cn/UG290E.pdf>
3. Power input 5V